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IG-APSO-DNN: Deep learning intrusion detection model to detect false data injection attacks in smart grids

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A R T I C L E I N F O

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A B S T R A C T

False Data Injection Attacks (FDIAs) present a significant threat to smart grids by manipulating measurement data, which may lead control centers to make incorrect operational decisions. Accurate and efficient detection of FDIAs is critical for ensuring reliable grid operation. Existing deep learning approaches often fail to capture both short-term local features and long-term dependencies in power grid data, and they typically show weak correlations with past and future time series information, reducing the trustworthiness of detection results. Similarly, conventional Intrusion Detection Systems (IDS) struggle to detect advanced FDIAs due to their reliance on predefined signatures and rule-based mechanisms. To overcome these limitations, we propose IG-APSO-DNN, a two-stage deep learning model for detecting FDIAs in smart grids. The first stage employs Information Gain (IG) and Adaptive Particle Swarm Optimization (APSO) for feature selection, reducing data dimensionality and improving model efficiency. The second stage uses a Deep Neural Network (DNN) to effectively capture both spatial and temporal patterns in smart grid measurements. The proposed model is evaluated on the Industrial Control System (ICS) Cyber Attack Power System Dataset, which simulates various FDIA scenarios. Results demonstrate that IG-APSO-DNN significantly outperforms traditional methods, improving key performance metrics including detection accuracy, precision, recall, and F-measure, while ensuring reliable operation of the smart grid. This study presents a robust anomaly-based IDS framework and highlights future directions, such as real-world validation, adaptive learning, exploration of novel optimization algorithms, and addressing scalability and real-time processing challenges.

1. Introduction

With the advancement of the Internet, artificial intelligence, and cyber-physical systems (CPS), traditional power grids have transformed into sophisticated smart grids. These modern systems integrate sensing, control, computing, communication, and decision-making capabilities into a unified infrastructure. In a typical CPS setup, the physical components of the power transmission network are closely interconnected with digital technologies. To maintain the security and stability of such smart grids, their optimization and standardization have become increasingly critical [1,2]. However, the widespread deployment of intelligent terminals and the reliance on data transmission over networks have introduced numerous vulnerabilities. As the intelligence of the grid grows, it also expands the potential surface for cyber threats. This heightened exposure makes smart grids susceptible to various cyberattacks, including replay attacks [3], denial-of-service attacks [4],

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topology manipulation [5], and False Data Injection Attacks (FDIAs) [6].

FDIAs pose a significant threat to the integrity of measurement data in smart grid Supervisory Control and Data Acquisition (SCADA) systems. These attacks may involve the deliberate alteration of sensor data, often by inserting carefully crafted false information into communication channels or sensor hardware. The manipulated data, once collected by SCADA from field devices, could mislead the control center during state estimation, potentially resulting in incorrect operational commands. Traditional Bad Data Detection (BDD) methods often fail to detect such sophisticated intrusions, thus increasing the risk of severe disruptions in power generation, transmission, and distribution. Consequently, effective detection and mitigation strategies are crucial for maintaining the stability and security of smart grid operations [7,8].

Intrusion Detection Systems (IDS) have become essential for safeguarding smart grids against cyber threats [9]. However, traditional IDS approaches, which rely on predefined rules and signatures, often struggle to detect sophisticated attacks like FDIAs [10,11]. To address this limitation, more adaptive solutions like anomaly-based IDS are being explored. Unlike signature-based methods, anomaly-based systems could identify deviations from normal behaviors, making them effective in detecting previously unseen attacks, including zero-day exploits and stealthy FDIAs. Furthermore, feature selection techniques could enhance IDS performance by isolating the most relevant features for FDIA detection, thus improving both detection accuracy and generalization across different attack scenarios [12,13].

In recent years, machine learning techniques have been increasingly applied in the development of network intrusion detection systems for smart grids, leading to notable improvements in effectiveness and performance [14–16]. However, many of these models struggle with large-scale datasets due to the high dimensionality and nonlinear nature of the feature space, which introduces significant computational complexity. This issue highlights the urgent need for more efficient dimensionality reduction algorithms to enhance the scalability and performance of IDS in smart grid environments.

Deep learning has emerged as a powerful approach for anomaly detection in smart grid systems, as it addresses the shortcomings of traditional rule-based methods that struggle with complex and evolving data patterns. Several IDS models, such as CNNs, DNNs, autoencoders, and RNNs can automatically extract meaningful features from raw sensor data and learn hierarchical representations to enable the detection of subtle and novel anomalies. Their ability to adapt to dynamic environments has enhanced the accuracy, flexibility, and efficiency of monitoring systems, thereby improving the security and reliability of smart grids and supporting broader applications across industrial, urban, and healthcare domains [17,18].

Current cybersecurity systems face significant challenges in managing the vast and complex data generated within interconnected smart grid environments, which could complicate the detection of sophisticated threats like FDIAs. Traditional feature selection techniques often fall short in identifying the most relevant features for accurate threat detection. Moreover, there is a noticeable gap in leveraging advanced bio-inspired optimization algorithms, such as the Adaptive Particle Swarm Optimization (APSO) and Information Gain (IG) for feature selection in smart grid security. Despite their proven potential, these methods remain underutilized. Existing optimization strategies for anomaly-based intrusion detection systems (AIDS) typically focus on single-objective criteria, which limit their effectiveness in real-world scenarios that demand multi-objective trade-offs [14]. The selection of Information Gain (IG) and Adaptive Particle Swarm Optimization (APSO) in this study is driven by their complementary capabilities in feature relevance evaluation and global optimization. IG is a well-established statistical measure that quantifies the contribution of individual features to class discrimination, thereby facilitating the elimination of redundant and irrelevant attributes while maintaining computational efficiency. Nevertheless, IG operates primarily as a univariate filter method and may overlook complex interdependencies among features. To mitigate this limitation, APSO is incorporated to optimize the selected feature subset through adaptive parameter adjustment and improved convergence dynamics. Compared with conventional Particle Swarm Optimization (PSO) or Genetic Algorithms (GA), APSO introduces adaptive control of inertia weight and acceleration coefficients, which enhances its ability to avoid premature convergence and strengthens global search capability in high-dimensional feature spaces [19]. The integration of IG and APSO thus establishes a hybrid feature selection framework that effectively combines relevance-based filtering with adaptive metaheuristic optimization, leading to more compact, discriminative, and robust feature representations for FDIA detection in smart grid environments. To address these issues, this study proposes a novel framework, known as the IG-APSO-DNN that combines IG and APSO for optimal feature selection, with a Deep Neural Network (DNN) classifier. This integrated approach could enhance detection accuracy and adaptability, thus offering a robust and scalable solution for defending smart grids against evolving cyber threats. The contributions of this study are outlined as follows:

I. This study has designed a feature selection model that could reduce the high dimension of data, while enhancing its efficiency using IG and APSO.
II. This study has designed an anomaly intrusion detection model that could detect attacks effectively in smart grids using DNN that utilizes feature selection (I).
III. The proposed IG-APSO-DNN model was validated using the Industrial Control System (ICS) Cyber Attack Dataset (Power System Dataset) that consisted of various FDIA scenarios simulated in a smart grid environment.
IV. The performance of the proposed IG-APSO-DNN model was compared with state-of-the-art.

The remainder of this paper is organized as follows: Section 2 reviews the related work, Section 3 presents the materials and methods, Section 4 details the proposed IDS, and Section 5 discusses the results and analyses. Finally, Section 6 concludes the paper.

2. Related works

Various techniques, including machine learning, game theory, attack graphs, Bayesian networks, and neural networks are used in the current research on FDIA detection systems [20–25]. The crucial contributions of FDI attack detection are emphasized in this section. Lakshmanan, et al. [26] proposed a Graph Neural Network-Transformer encoder model (GNN-Transformer) for anomaly detection in smart grids to detect FDIAs. This model used graph neural networks to extract spatial features and transformer encoders to capture long-range dependencies with high accuracy, which outperformed other models, such as the GMM and SPP-CNN. This model was evaluated using the PMU dataset and achieved high accuracy of 97.90 %, precision at 0.82, and recall at 0.98. Scholar [27] proposed a detection approach using the Adaptive Residual Recurrent Neural Network (ARRN) with the Dilated Gated Recurrent Unit (DGRU) for improved anomaly identification of FDIAs in a smart grid. This model was evaluated using the Electrical Grid Stability Simulated (EGSS) Dataset. Similarly, Li, et al. [28] proposed an intrusion detection model to address the changes in power grid topology using the gated graph neural network GGNN. To evaluate this model, they built their own dataset instead of using a benchmark dataset. This model achieved 95.13 % accuracy and 1.56 % for missing alarm rate.

Wang, et al. [29] proposed an FDIA detection model that leveraged both spatial and temporal features using the Graph Convolutional Networks (GCNs) to model spatial relationships across buses and the Temporal Convolutional Networks (TCNs) to capture sequential patterns. When tested on IEEE 14-bus and 118-bus systems, their model showed high accuracy and robustness. The results demonstrated that this model exhibited strong detection accuracy and robustness. Similarly, Li, et al. [30] introduced a Spatial-Temporal Transformer Network (STTN) that combined graph convolution with self-attention for spatial modeling and attention mechanisms for capturing long-range temporal dependencies to address limitations of traditional LSTMs. Their method achieved 92.59 % accuracy on real NYISO load data applied to IEEE 14-bus and 118-bus systems.

Takiddin, et al. [31] tackled the vulnerability of FDIA detectors to data poisoning caused by mislabeled training samples. They proposed a robust anomaly detector, known as the CR-GAE (Convolutional Recurrent Graph Autoencoder), trained solely on benign data. It combined the Chebyshev graph convolutions, recurrent layers, and attention mechanisms to capture spatiotemporal dependencies. Evaluations on IEEE 14-, 39-, and 118-bus systems showed strong detection performance, with only 1.6 %–3.7 % degradation, even under severe poisoning and unseen topologies. On the other hand, He, et al. [32] introduced the A-BiTG model, which integrated a bidirectional TCN, BiGRU, and attention mechanism to detect FDIAs. It effectively captured both past and future temporal dependencies, with the attention component highlighting critical features. Tests on IEEE 14- and 118-bus systems demonstrated its resilience and high accuracy under various noise levels and attack intensities.

Abudin, et al. [33] investigated the impact of FDIAs on smart grids using data from a 10 kV solar PV system in a lab setting. They evaluated several machine learning models, such as the Random Forest, Isolation Forest, Logistic Regression, Decision Tree, Autoencoder, and Feedforward Neural Network for FDIA detection. Their study has highlighted both the effectiveness of these models in identifying anomalies and the critical role of machine learning in enhancing smart grid security. In a similar manner, Pasumponthevar and Jeyaraj [34] proposed a hybrid intrusion detection framework (KFRNN) by combining Kalman filters and recurrent neural networks. Their model incorporated adaptive feature boosting and reinforcement learning via a stacking strategy. The Kalman filters reduced feature dimensionality and improved the robustness of the model. Tested on the WUSTIL-2021 dataset and a HIL testbed, their model achieved 97.3 % accuracy, and demonstrated strong performance and resilience in additional IEEE 118-bus system evaluations.

Despite significant advancements in FDIA detection using deep learning, graph-based, and hybrid models, several critical research gaps remain. Many existing approaches rely heavily on fully labeled, clean datasets, which could make them vulnerable to mislabeled data and poisoning attacks. While some models could effectively capture either spatial or temporal dependencies, few offer comprehensive solutions that generalize well across diverse grid topologies and unseen attack scenarios. Moreover, real-world validation is often lacking and adversarial robustness is insufficiently explored, further limiting practical deployment. These challenges underscore the need for more robust, generalizable, and label-independent models that are capable of operating reliably in dynamic and adversarial environments. A summary of the reviewed approaches is presented in Table 1.

Table 1
Summary of recent FDIA detection models, including techniques used, datasets, performance results, and identified limitations.

Paper	Year	Techniques used	Dataset	Results	Limitations
[26]	2024	GNN & Transformer encoder	PMU	97.90 %	-High dimensional data not addressed. -High computational complexity. -FAR not reported
[27]	2024	ARRNN & DGRU	EGSS	0.94 %	-High dimensional data not addressed. -High computational complexity.
[28]	2024	GGNN	Test-bed	95.13 %	-High dimensional data not addressed. -Precision, recall, fi-score evaluation matrix not reported.
[29]	2025	GCN & LSTM	NYISO	95.88 %	-High dimensional data not addressed. -High computational complexity.
[30]	2024	STTN & self-attention	NYISO	92.59 %	-Low detection accuracy.
[31]	2023	CR-GAE	Test-bed	95.9 %	-High dimensional data not addressed. -High computational complexity.
[32]	2024	A-BiTG	NYISO	92.67 %	-Low detection accuracy. -High dimensional data not addressed. -High computational complexity.
[33]	2024	Autoencoder	Test-bed	95 %	-Low detection accuracy. -High dimensional data not addressed.
[34]	2024	KFRNN	Test-bed	97.3 %	-False data injection attacks can disrupt state estimation results.

3. Materials and methods

The accumulation of high-dimensional gene expression data has a substantial influence on deep learning models, since they could increase the level of computational complexity and the possibility of overfitting. Thus, this study proposes a hybrid feature selection framework that incorporated IG and APSO, along with a Random First initialization technique. This framework was intended to help overcome the previously discussed issues. The first step taken in this study was to perform statistical filtering and ranking. At the outset, low-variance characteristics were eliminated by employing a variance threshold to get rid of uninformative traits. Next, the remaining features were standardized and sorted using IG. This method was used to quantify the degree of reliance that existed between each feature and the target class. A smaller and more relevant feature space was produced as a result of the elimination of features that have IG scores that were lower than the predetermined threshold. The APSO algorithm was utilized in the second step to identify the most suitable subset of the remaining features. A Random First initialization was adopted, which in contrast to the usual PSO, encouraged early variation in the particle population. This, in turn, enhanced the exploratory capability of the search. After each cycle, the APSO performed a dynamic adjustment to the weight of its inertia to achieve an optimal balance between exploration and exploitation. The goal function ensured a compact and high-performing selection by combining a penalty for bigger feature subsets, with a classification accuracy evaluation that was performed using a Random Forest classifier. An early stopping mechanism was also incorporated, which resulted in the optimization process being terminated, if after a number of iterations, no discernible gain in fitness was measured. By identifying a small subset of highly informative features in an efficient manner, this adaptive technique was able to dramatically improve the performance of the model, while simultaneously minimizing the amount of time and complexity required for training.

3.1. Information gain (IG)

Information Gain (IG) calculates the value of each feature and selects those whose IG value exceeds a predefined threshold to form the selected feature subset [35]. A large IG value is indicative that the feature contains more information relevant to the target class. Generally, the higher the IG value of a feature, the greater its discriminative power. To enhance the relevance and efficiency of the proposed classification model, a filter-based feature selection method utilizing IG was employed. Initially, features exhibiting low variance were removed, as they contributed minimally to distinguishing between classes. Specifically, features with variance below a threshold of 0.01 were excluded. The remaining features were then standardized using Z-score normalization by transforming each feature to have a zero mean and unit variance. This normalization ensured that all features would contribute equally when computing statistical measures. Following standardization, IG was used to quantify the relevance of each feature, with respect to the target class. IG, also referred to as mutual information in the context of feature selection, measured the reduction in uncertainty about the class labels after observing a feature [36]. Features with IG values that exceeded a predefined threshold of $\theta = 0.01$ were selected for further processing. This filtering process was used to eliminate irrelevant or redundant features, thereby improving the computational efficiency and interpretability of this model. To visualize the results, the IG scores were plotted in a descending order to reveal the most informative features. This entropy-based approach ensured that only features that were most strongly associated with class discrimination were retained in the final dataset.

The detailed steps of the IG-based feature ranking and grouping approach are outlined in Algorithm 1.

Require: Feature set: F={ f_1, f_2,…, f_m }
IG threshold θ_{IG}
Ensure: Selected feature subset $F_{selected} \subseteq F$
FOR i=1, 2, …,m do
1.Compute the entropy H (f) of the target class according to Equation (2)
2.Compute the conditional entropy H (f /S) of each feature according to Equation (3)
3.Compute the Information Gain (IG) of each feature using Equation (1):
END FOR
4.Sort the features in descending order according to their IG values to obtain a ranked feature set:
IG (f) = {IG (f_1), IG (f_2), …., IG (f_m)}
5.Initialize $f_{selected} \leftarrow \varnothing f$
FOR i=1, 2, …,m do
6.**IF** IG (f) θ_{IG} **THEN**
Add f_1 to $f_{selected}$
END IF
END FOR
return Selected features: $f_{selected}$

3.2. Adaptive particle swarm optimization (APSO)

Particle swarm optimization (PSO) is an evolutionary algorithm inspired by the social behavior of a flock of birds [37]. It has found applications in a myriad of optimization problems, among which is feature selection in machine learning. In PSO, each particle denotes a potential solution in the high-dimensional solution space. In combination with each other, they would go in the direction of the optimal solution by modifying the position of the given particle based on individual and global best experiences. Unfortunately, traditional PSO suffers from harsh convergence, resulting in suboptimal performance for complex, high-dimensional data. APSO was used in this study to overcome this limitation, which dynamically adjusted the inertia weight according to the fitness values of the swarm. An inertia weight is crucial in balancing global exploration into a local exploitation of the searched space. Every time it iterates, APSO changes this weight and thereby de-stagnates the local optima and promotes convergence to the optimal feature subset.

This study integrates APSO for feature selection within an IoT-based intrusion detection system, following the structured workflow presented in Fig. 1. The algorithm began by initializing a population of particles, where each particle represented a subset of features. Each particle has a position (x_i^k) and velocity (v_i^k) at iteration k_i, represented as $x_i^k = [x_{i1}^k, x_{i2}^k, ..x_{id}^k]$ and $v_i^k = [v_{i1}^k, v_{i2}^k, ..v_{id}^k]$, respectively. These particles were then evaluated based on a classification performance metric, specifically accuracy, using a machine learning model. The best-performing particles updated their personal best (pBest) and global best (gBest) scores to

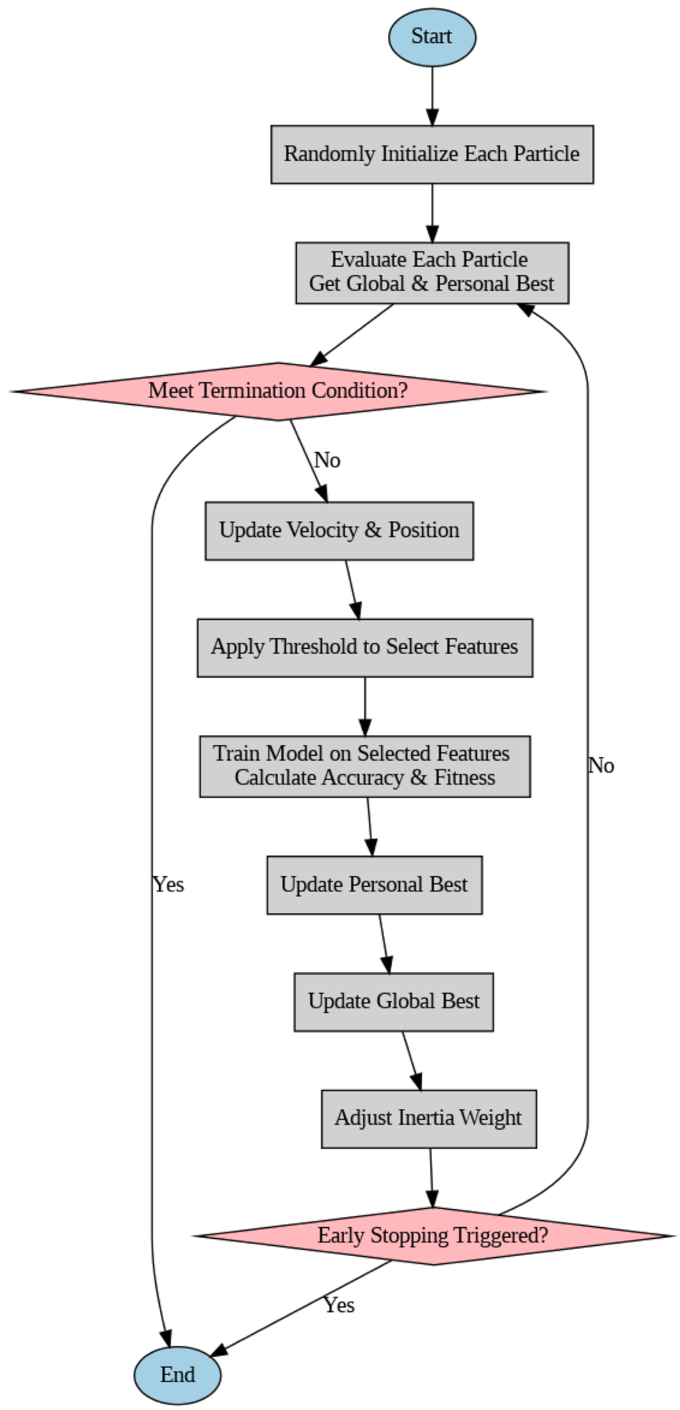

Fig. 1. The proposed APSO.

guide the optimization process. The velocity update was governed by the following Eq. (4):

$$v_i^{k+1} = wv_{id}^k + c_1 r_1 (pBest_{id} - x_{id}^k) + c_2 r_2 (gBest_{id} - x_{id}^k) \quad (1)$$

where w is the inertia weight, c_1 and c_2 are the cognitive and social coefficients, while r_1 and r_2 are random numbers uniformly distributed in [0,1]. The updated velocity was then used to modify the position of the particle a in Eq. (5):

$$x_{id}^{k+1} = x_{id}^k + v_{id}^{k+1} \quad (2)$$

The position and speed were updated using both cognitive and social components of particle movement to find a better way to select features with a balanced exploration versus exploitation option. In addition, a threshold was applied to enhance selection by removing unimportant features from the analysis. The fitness function was determined by classification accuracy, thus, the main evaluation considered for assessing how well a given feature subset performed was based on classification accuracy. It was used to evaluate which features were important for analysis. To further improve optimization efficiency, the adaptive inertia weight was dynamically adjusted based on the fitness distribution of the population to prevent premature convergence and to enhance performance. The adaptive inertia weight was computed using Eq. (6):

$$\omega = \begin{cases} \omega_{max}, \ if \int_{cur} > \int_{avg} \\ \omega_{min} - (\omega_{max} - \omega_{min}).\dfrac{\int_{cur} - \int_{min}}{\int_{avg} - \int_{min}}, \ if \int_{cur} \leq \int_{avg} \end{cases}. \tag{3}$$

where $\int_{cur}$ is the fitness value of the current particle, $\int_{avg}$ is the average fitness of the population, and $\int_{min}$ is the minimum fitness in the population.

The loop continued until a certain termination condition has been fulfilled, for example, reaching the maximum iterations or satisfying the convergence criteria, thus, producing an optimized subset of features for intrusion detection. By using APSO, the feature selection technique was refined to increase classification accuracy, while simultaneously reducing computational complexity. This approach ensured that only the most pertinent features were preserved for downstream model training. The pseudocode of APSO used in this study is as shown in Algorithm 2. APSO implementation in this study introduced two key enhancements. First, it employed parallel processing via the Joblib Parallel module to evaluate the fitness of particles concurrently, thereby significantly reducing computation time when handling high-dimensional feature spaces and complex models. Second, strict binary position enforcement was applied after velocity updates and binary thresholding to ensure that each particle could consistently represent a valid feature subset. Table 2 shows the hyperparameters of the proposed APSO.

Algorithm 2. The pseudocode of the proposed APSO.

```
Initialize the number of particles and iterations
Initialize the inertia weight (w), cognitive (c1), and social (c2) coefficients
Set velocity limits (min_velocity, max_velocity)
while i < num_particles do
```

(continued on next column)

(continued)

```
    Initialize the position Xi and velocity Vi of each particle
    Calculate the fitness value of each particle fitness(i)
    i = i + 1
end
Initialize pBesti = Xi
Initialize gBest = max(pBesti) based on fitness
for g = 1 to num_iterations do
    for i = 1 to num_particles do
        Update Vi and Xi using Equations (2) and (3)
        Calculate the new fitness value fitness(i)
        if fitness(i) > fitness(pBesti) then
            pBesti = Xi
        end if
        if fitness(i) > fitness(gBest) then
            gBest = Xi
        end if
    end for
    Calculate the fitness distribution ∫cur, ∫avg, ∫min
    Adjust inertia weight (w) dynamically using Equation (4)
    Adjust c1 and c2 adaptively
    if evolutionary state S == convergence state S3 then
        σ = σmax − (σmax − σmin) × g / num_iterations
        Td = Td + (Xdmax − Xdmin) × Gaussian (μ, σ^2)
        if fitness (Td) > fitness(gBest) then
            gBest = Td
        end if
    end if
end for
Return the best feature subset from gBest
Train a final model using the selected features and evaluate accuracy
```

Table 2
The APSO hyperparameters.

Hyperparameter	Value	Description
Number of Particles	10	Number of particles in the swarm.
Number of Iterations	5	Maximum number of iterations for optimization.
Inertia Weight (w)	0.9 (initial), adaptive	Controls balance between exploration and exploitation, decreasing over iterations.
Cognitive Weight (c1)	1.5	Influence of personal best position on velocity update.
Social Weight (c2)	1.5	Influence of global best position on velocity update.
Minimum Velocity	−0.1	Lower bound for velocity updates.
Maximum Velocity	0.1	Upper bound for velocity updates.
Fitness Function	Accuracy	Evaluates feature subsets using a Random Forest classifier.
Feature Representation	Binary 0 or 1	Each particle represents a binary feature selection vector.
Adaptive Inertia Weight	Decreases over time (0.99 factor)	Helps prevent premature convergence.
Parallel Processing	Enabled (n_jobs=−1)	Utilizes multiple cores for faster evaluation.
Final Classifier	Random Forest (50 trees)	Trained on the best feature subset for final evaluation.

3.3. Classification model using DNN

Deep Neural Networks (DNNs) have also been employed for general classification tasks, since their underlying philosophy empowers them to learn hierarchical representations from raw data [38,39]. DNNs are a class of artificial neural networks that are characterized by multiple hidden layers between the input and output layers, which enable them to learn complex representations from high-dimensional data. Fundamentally, each neuron in a DNN computes a weighted sum of its inputs, followed by a non-linear activation. Mathematically, the transformation of each layer can be represented by the following Eqs. (7) and 8:

$$z = w.^{x+b} \tag{4}$$

$$a = \int (z) \tag{5}$$

where w represents the weight, b is the bias, x is the input vector, z is the linear combination of inputs, and $\int(z)$ is an activation function (for example, ReLU, in this case: $\int(z) = \max(0, z)$). The layered structure of a DNN makes it competent to work on intrusion detection tasks by enabling automatic feature extraction hierarchically from raw data. The DNN used in this study was an intrusion-detection-targeted network for IoT type of environments and it used an optimized feature subset from APSO. The DNN architecture comprised multiple fully connected layers interspersed with dropout layers. This technique has been demonstrated to reduce overfitting effectively [40]. In this study, dropout was applied at the rates of 0.3 and 0.2 in the first and second hidden layers, respectively, to enhance model generalization. This model was optimized using the Adam optimizer, known for its adaptive learning rate and efficient convergence properties. These design choices, combined with the APSO-based feature selection, contributed to a robust, accurate, and lightweight IDS, as supported by prior studies in this field. Fig. 2 shows the architecture of the proposed DNN classification model.

4. The proposed IDS

Fig. 3 presents the whole workflow of the proposed IG-APSO-DNN-

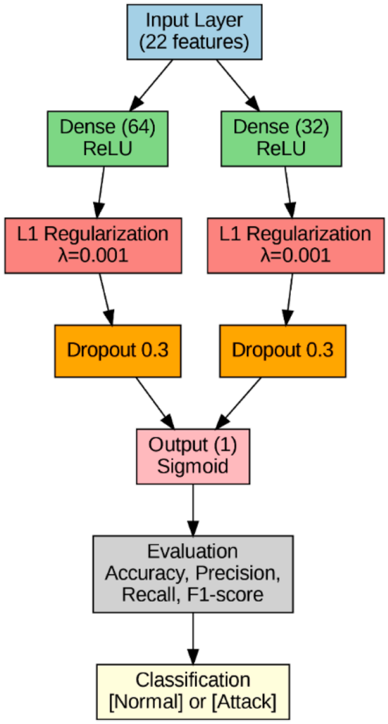

Fig. 2. The proposed DNN model.

based intrusion detection system (IDS). This IDS was developed primarily to identify FDIAs in smart grid environments. Data cleansing, labelling, and normalization of the benchmark dataset were all part of the complete data preprocessing steps that the framework was built upon. This phase was designed to ensure that the data were of high quality and consistent throughout. Subsequently, this system conducted hybrid feature selection by integrating IG and APSO to find and keep the most informative features. This step helped reduce the dimensionality of the model and improved its efficiency. Next, the selected characteristics were fed into a lightweight deep neural network classifier that has been customized to capture the temporal and spatial dependencies that are inherent in smart grid data. The suggested deep neural network design was purposefully kept at a low level, consisting of only two completely dense connected layers. This was done with the objective of preserving a balance between detection performance and processing economy. Based on the held-out test dataset, the final part of the evaluation process involved evaluating the performance of the IDS by utilizing important classification measures. These metrics included accuracy, precision, recall, F1-score, and the area under the curve (AUC). The fundamental purpose of the IG-APSO-DNN framework was to offer good detection performance for FDIAs, while simultaneously minimizing computational overhead. These characteristics would allow the framework to be suited for real-time smart grid applications.

4.1. Experimental setup

The proposed IG-APSO-DNN model was implemented using a combination of hardware and software platforms. The hardware included a Toshiba laptop with an Intel® Core™ i5 CPU M 460 @ 2.53 GHz and 8 GB of RAM, as well as a Kaggle standard notebook for additional computational support. The software environment consisted of Python 3.10, with key libraries including TensorFlow 2.12 for deep learning, NumPy 1.24 for numerical computations, Pandas 2.0 for data handling, and Scikit-learn 1.2 for feature preprocessing and evaluation metrics.

4.2. Dataset

This research employed a benchmark Power System Dataset, which contained realistic smart grid network traffic to evaluate the effectiveness of the proposed IG-APSO-DNN model and other comparative models. This dataset has been widely used in related studies for the evaluation of IDS in smart grid environments [41,42]. The Power System Dataset was generated using a realistic power system testbed that emulates the operation of an electrical grid. The setup includes physical and simulated components such as a generator, transmission lines, circuit breakers, and protective relays, all controlled through Modbus/TCP communication. Network traffic was captured during both normal and attack scenarios, providing comprehensive coverage of real-world smart grid operations. The dataset contains multiple types of cyberattacks, particularly False Data Injection (FDI) attacks, which manipulate measurement data to mislead the state estimation process of the grid. Other simulated attack types include Denial-of-Service (DoS), Replay, and Command Injection attacks. However, for this study, we focused on the FDI attack instances to evaluate the effectiveness of the proposed IG-APSO-DNN model in detecting stealthy data manipulation behaviors within smart grid environments. Each record in the dataset includes 129 features representing various electrical and communication parameters (e.g., voltage, current, frequency, phase angle, breaker status, and Modbus packet attributes). The data were preprocessed to remove redundancy and balance class distribution before model training and evaluation. Table 3 shows the distribution of FDI and Non-FDI records in this dataset.

This benchmark dataset provides a realistic foundation for evaluating IDS performance and assessing model robustness in smart grid cybersecurity applications.

4.3. Dataset preprocessing

This stage involved preparing the raw benchmark dataset for analysis, which was conducted in three main steps, as described in the following subsections.

4.4. Cleaning

In this stage, the dataset underwent a thorough cleaning process to remove noise, irrelevant information, and potential sources of error, such as missing values and outliers. It was essential to ensure data integrity at this point to enhance the quality and dependability of the subsequent analysis. This cleaning phase targeted incomplete, inconsistent, or distorted records that could negatively impact model accuracy. Additionally, features that did not contribute meaningful variations, for example, those containing identical values across all records, were excluded to reduce model complexity and computational overhead [43]. To be more specific, any attribute that contained null entries or variables that remained constant throughout the dataset was eliminated. The subsequent refined and validated dataset served as the input for the following labelling procedure.

4.5. Labeling

This phase involved assigning class labels to the dataset, which was a critical step in preparing data for supervised learning tasks. In supervised machine learning, labeled data are essential for training models to distinguish between predefined categories, whereas unsupervised approaches operate on unlabeled data [44]. The benchmark dataset used in this study originally lacked explicit class labels. Therefore, during this stage, the data were annotated to reflect two categories: i) FDIA (1) that indicated the presence of a False Data Injection Attack, and ii) No FDIA (0) that indicated its absence. Labeling is fundamental for evaluating intrusion detection systems, which depend on clearly defined categories to identify and differentiate between normal and malicious traffic. Thus,

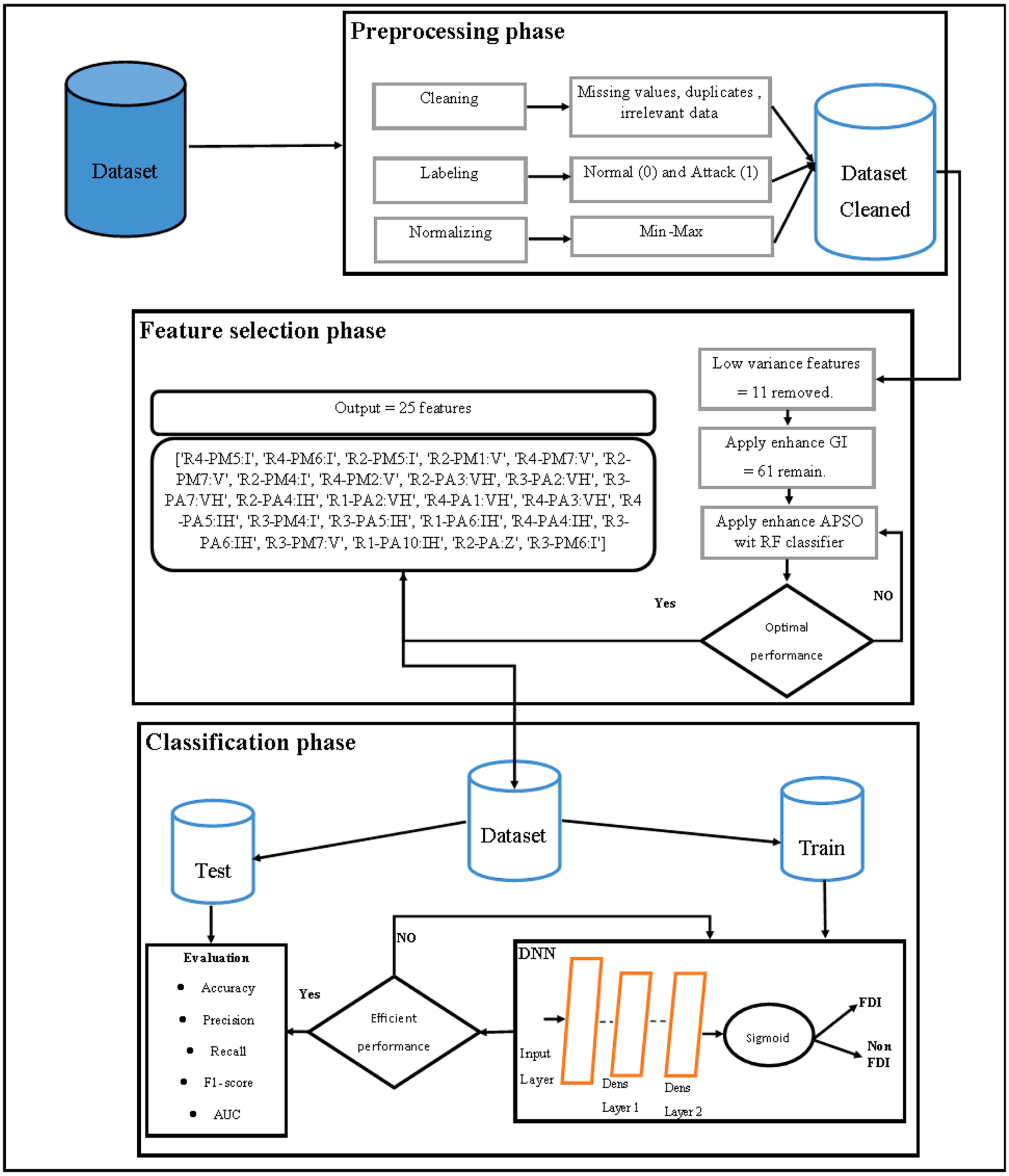

Fig. 3. The proposed IG-APSO-DNN model.

Table 3
The distribution of FDI and Non-FDI in the Power System Dataset.

Dataset	Features	Non-FDI	FDI
	129	136,284	944,36
Total		230,720	

all records in the dataset used in this study were systematically labeled to support accurate model training and evaluation.

4.6. Normalization

To mitigate the effects of high dimensionality and to enhance the accuracy of machine learning models, data normalization is considered a vital preprocessing step. As highlighted by Sinaga et al. [45], normalization plays a key role in ensuring that input features contribute equally to the learning process, particularly when they originate from different scales or units of measurement. One widely used normalization technique is the StandardScaler, which transforms features to have a mean of 0 and a standard deviation of 1. This transformation helps eliminate scale-induced bias and improves model stability. For certain use cases, min-max normalization may also be employed to rescale numerical features within a fixed range of commonly [0, 1]. This transformation is expressed mathematically in Eq. (9):

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{6}$$

where X represents the feature value, while X_{min} and X_{max} denote the minimum and maximum values of that feature, respectively. This normalization ensures that all features are scaled consistently, making the data more suitable for analysis and reducing the impact of disproportionate value ranges [46]. In this study, normalization was applied uniformly across training, validation, and testing phases to maintain consistency and to optimize the performance of the proposed classifier.

4.7. Evaluation matrix

To evaluate the effectiveness of the IDS, a range of performance metrics were employed. Among the most widely used metrics are accuracy, false positive rate (FPR), and false negative rate (FNR). These metrics are commonly derived from the confusion matrix, which is particularly suited for binary classification tasks. A confusion matrix is a tabular representation that summarizes the classification outcomes, where the columns correspond to the predicted classes and the rows correspond to the actual classes. The TP (True Positives) represents correctly identified attack instances, TN (True Negatives) refers to correctly identified normal traffic, FP (False Positives) indicates normal traffic mistakenly classified as an attack, and FN (False Negatives) refers to attacks that are incorrectly classified as normal traffic. These values provide the foundation for calculating various performance indicators that reflect the accuracy, reliability, and ability of the IDS to minimize both false alarms and missed detections. A detailed summary of these results is presented in Table 4.

The following Equations 10 to 14 were used to evaluate the performance of the proposed IDS [39–47]:

$$Ac = \frac{TP + TN}{FP + FP + TN + TN} \tag{7}$$

$$\text{Recall } = \frac{TP + TP}{TP + FN} \tag{8}$$

$$\text{Precision } = \frac{TP}{TP + FP} \tag{9}$$

$$\text{FAR } = \frac{FP}{TN + FP} \tag{10}$$

$$\text{F1 } = 2.\frac{\text{precision} \cdot \text{recall}}{\text{precision } + \text{ recall}} \tag{11}$$

5. Results and discussion

In the initial phase of the intrusion detection framework, feature selection was employed to reduce dimensionality and to enhance the interpretability and efficiency of the detection model. This process began by removing low-variance features, which were unlikely to contribute significantly to the classification task. Specifically, 11 features with variance below the threshold of 0.01 were eliminated. These features included redundant or static signals, such as 'R1:DF', 'snort_log1′, 'control_panel_log1′, and other similar logs from control panels and routers. Table 5 shows the low variance features in the dataset.

Normalization was performed on the remaining dataset after the low-variance features have been removed. This removal was done to guarantee that all attributes were represented on a uniform scale, which made it easier to conduct an evaluation that was both fair and balanced

Table 4
CONFUSION matrix.

Sample class	Prediction		
	Normal	Attack	
Real	Normal	TP	FP
	Attack	FN	TN

Table 5
The low variance features in the dataset.

Threshold	Low variance features
0.01	['snort_log3′, 'R2:DF', 'control_panel_log4′, 'control_panel_log1′, 'snort_log1′, 'R4:DF', 'control_panel_log3′, 'snort_log2′, 'snort_log4′, 'R1:DF', 'control_panel_log2′]

regarding the value of the features. Then, IG scores were estimated using mutual information, with regard to the binary class label that differentiated between occurrences of regular data injection and instances of FDI. A quantifiable indicator of the contribution that each feature made to the classification task was provided by the identification metric. A significance criterion of 0.01 was given to the IG scores to refine the feature space after it had been analyzed. It was determined that features with scores lower than this level did not possess sufficient discriminatory power, and as a result, they were cut out of consideration. Consequently, 59 of 117 features that were initially present were kept, which resulted in a reduction in the dimensionality by approximately 50 %. The decreased dimensionality has a significant role in reducing the amount of computing overhead and lowering the danger of overfitting in future learning techniques. A total of 47 features were eliminated because of their low IG scores. These features generally consisted of a variety of sensor readings, including voltage, current measurements, and log data, which exhibited negligible significance to the identification of malicious behaviors. Notable among these were attributes, such as R3:F, R1-PM3:V, R4-PM12:I, and relay3_log. These attributes displayed a limited amount of mutual information with the target variable, and as a result, they were seen as redundant in the context of intrusion detection.

IG filtering was able to drastically reduce the number of redundant features; nevertheless, it failed to identify the ideal feature subset in relation to the performance of the proposed model. APSO was then applied to the feature set to further refine the selection. Therefore, the objective of this metaheuristic technique was to establish an ideal trade-off between detection accuracy and computing efficiency. This objective was accomplished by balancing the importance of features and the performance of models, while penalizing the size of feature sets. The settings that allowed the APSO algorithm to alter the inertia weights in an adaptive manner between rounds were used to initialize the algorithm. This was done to encourage both exploration and exploitation. To identify which features should be included, a binary mask representation was utilized. Additionally, a dynamic threshold that was based on the inertia weight was utilized to guarantee adaptive feature selection across the iterations. To encourage this model to discover minimum, yet highly informative feature subsets, the fitness function was specified as the sum of the classification error, which was equal to one minus the accuracy, and a penalty that was proportional to the number of features that were selected. For the purpose of evaluating the fitness function, a Random Forest classifier that contained 300 estimators was utilized because of its resilience and ability to accurately represent tabular datasets. To identify the optimal settings, APSO was executed multiple times under varying configurations of swarm sizes and iteration limits. As shown in Table 6, varying the number of particles and iterations has

Table 6
Optimization Results of IG-APSO with Different Particle and Iteration Configurations.

Number of particles	Number of iterations	Selected features	Time/ sec.	Fitness	Accuracy
5	10	25	539.7931	0.3255	0.93
5	15	27	592.1224	0.3421	0.93
10	15	31	582.9389	0.3848	0.92
10	10	31	400.4523	0.3864	0.92
10	20	27	795.7435	0.3457	0.92
5	20	29	836.5472	0.3623	0.92

yielded different trade-offs between feature count, execution time, and classification accuracy.

With a swarm size of five and 10 iterations, the optimal performance was reached among these setups. This resulted in a fitness score of 0.3255, a classification accuracy of 93 %, and the selection of only 25 features. Additionally, this arrangement displayed efficient convergence, with early stopping conditions being reached in certain instances where no discernible progress was detected across numerous rounds. Specific characteristics, such as R4-PM6:I, R2-PM5:I, R2-PA4:IH, and R4-PA3:VH were highlighted among the chosen features. These characteristics encompassed a wide range of sensor types and devices that were part of the smart grid. It was highly probable that these characteristics were suggestive of aberrant patterns and disturbances that could be linked to FDIAs, which may suggest that they played an important role in the detection of intrusions. Fig. 4 illustrates the convergence behavior and model performance throughout different iterations. This figure demonstrates that there is a steady improvement in fitness, while there is a minor decrease in classification accuracy. It is crucial for real-time, resource-constrained smart grid applications that the APSO framework could successfully strike a balance between accuracy and feature reduction. Although the predictive performance of the proposed model was maintained or even improved, the dimensionality of the features was greatly decreased by the use of APSO-based feature selection following the IG filtering process. This observation illustrated that adaptive swarm intelligence was useful in effectively optimizing feature subsets for the purpose of ensuring the security of cyber-physical systems. The feature selection approach that was utilized in this study, known as the IG-APSO, has not only made the dataset easier to understand, it could also guarantee that only the most interesting and informative variables were kept. This ability would be especially beneficial in situations where there is a limited number of computational resources available, such as scenarios involving real-time or edge-based deployment environments. These findings provided further evidence that a robust feature selection method could play a significant role in improving the efficacy, interpretability, and efficiency of intrusion detection models for smart grid systems. To summaries, the IG-APSO pipeline has not only decreased the dimensionality of the original dataset by >75 %, it has also maintained a high level of detection accuracy without sacrificing precision. The resultant tiny feature subset was particularly well-suited for use in real-time smart grid IDS.

After applying the IG-APSO feature selection method, the dataset was divided into training and testing subsets to enable supervised learning. The final split included 184,576 samples for training and 46,144 samples for testing, thus preserving class balance to ensure an unbiased evaluation. Table 7 shows the dataset distribution after the split.

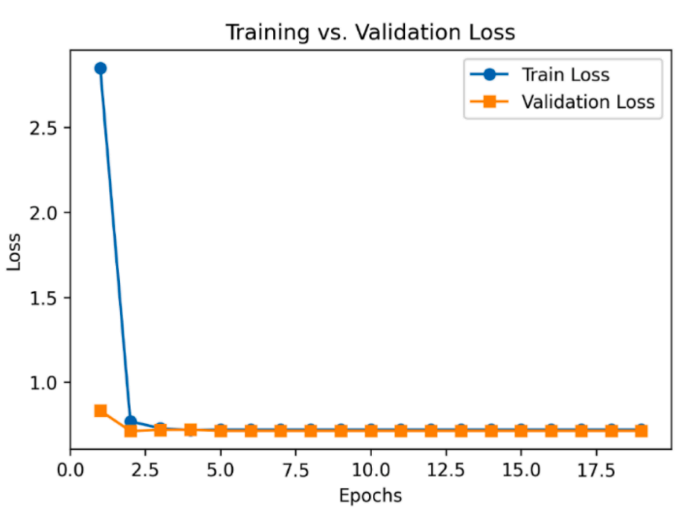

Fig. 4. Training and validation loss of the proposed GI-APSO-DNN model.

Table 7
Dataset distribution after the spilt.

Dataset	Data	Records
	X_train	184,576
	X_test	46,144
	y_train	184,576
	y_test	46,144

A DNN was developed to classify between normal and FDI instances. The network input consisted of 25 features selected through the IG-APSO pipeline. The hyperparameters of the proposed IG-APSO-DNN were determined through extensive experimentation, where the model was trained with different configurations such as hidden layer sizes of [32,16], [64, 32], and [128, 64], dropout rates ranging from 0.2 to 0.5, and learning rates from 0.0001 to 0.001. Among these, two hidden layers with 64 and 32 neurons, a dropout rate of 0.3, and a learning rate of 0.0001 provided the best balance between classification performance and generalization. These settings allowed the network to effectively learn from the 25 selected features while avoiding overfitting. The architecture is as previously shown in Fig. 3. Table 8 shows the hyperparameters of the proposed IG-APSO-DNN.

Fig. 4 illustrates the training and validation accuracy over the epochs, while Fig. 5 displays the training and validation loss curves. The accuracy curves revealed a steady increase and convergence, with training accuracy reaching 97.5 % and validation accuracy stabilizing at 96.8 %, while the FAR was at 2.11 %. The corresponding loss curves demonstrated consistent convergence, with no significant divergence, indicating minimal overfitting and a well-regularized model.

To assess the model's generalization, the DNN was evaluated on the holdout test set comprising 46,144 samples. Fig. 6 shows the proposed GI-APSO-DNN's performance, which achieves accuracy of 97.90 %, **precision** of 98.50 %, recall of 97.90 % and F1-score of 98.20 %. Table 9 shows the classification report.

The confusion matrix provided further insight into the model's performance. As shown in Fig. 7, the majority of the samples have been correctly classified, with minimal false positives and false negatives. These results indicated that the proposed model could effectively reduce misclassification errors, thus enhancing the reliability of the IDS.

The Receiver Operating Characteristic (ROC) curve in Fig. 8 demonstrates the model's ability to differentiate between normal and anomalous traffic. This figure also shows that the area under the curve (AUC) value of the ROC_Curve is 0.986, which further supports the high discrimination capability of this model.

To rigorously evaluate the efficacy of the proposed IG-APSO-DNN model, a comparative analysis was conducted against several state-of-the-art models reported in recent literature. This evaluation was based on four standard classification metrics: Accuracy, Precision, Recall, and F1-Score. These metrics collectively provided a comprehensive assessment of model performance across both detection reliability and robustness. As shown in Table 10 and Fig. 9, the proposed IG-APSO-DNN

Table 8
DNN hyperparameters.

Hyperparameter	Value / Description
Input features	25 (selected by IG-APSO)
Number of hidden layers	2
Hidden layer sizes	64 and 32
Activation function	ReLU
Output layer	1 neuron (sigmoid activation)
Dropout rate	0.3
Optimizer	Adam
Learning rate	0.0001
Loss function	Binary Cross-Entropy
Batch size	64
Number of epochs	10

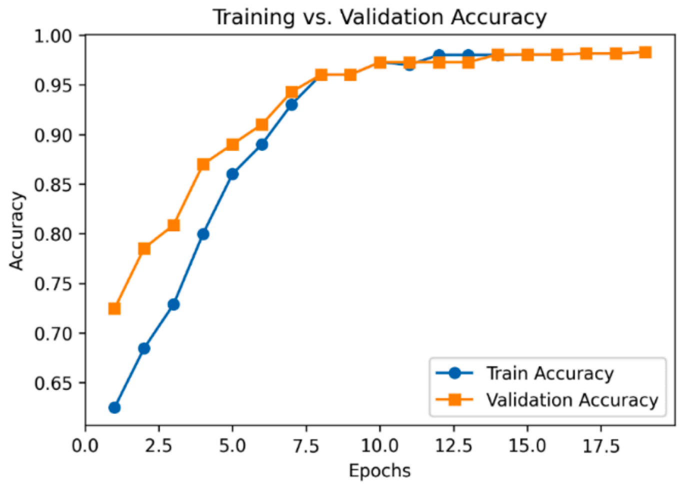

Fig. 5. Training and validation accuracy of the proposed GI-APSO-DNN model.

model achieves the highest performance across all evaluation metrics, demonstrating its superior classification capability. Specifically, it attained an accuracy of 97.90 %, outperforming the most competitive baseline model, the GCN-LSTM, by Wang, et al. [29], which achieved 95.88 % accuracy. This improvement underscored the effectiveness of the proposed model in capturing complex patterns within the data. In terms of precision, the proposed model achieved a score of 98.50 %, which was slightly lower than the precision percentage reported by Wang, et al. [48] at 99.60 %. However, it is important to note that the DAE's high precision was counterbalanced by a significantly lower recall of 88.60 %, indicating a tendency to miss a substantial portion of relevant instances. In contrast, the proposed IG-APSO-DNN model maintained a balanced performance, with both precision and recall at 98.50 % and 97.90 %, respectively. This balanced trade-off would be critical in anomaly detection scenarios, where both false positives and false negatives carry significant operational risks.

The F1-score, which harmonized precision and recall, further highlighted the robustness of the proposed approach. The proposed model recorded an F1-score of 98.20 %, surpassing all comparative methods, including the MFDA-ARRNN-DGRU by Scholar [27] and the DAMGAT by Wang, et al. [49], which achieved 94.03 % and 93.42 %, respectively. These results indicated the proposed model's superior ability to maintain consistency and effectiveness across varied data distributions. These observed performance gains could be attributed to the integration of IG for discriminative feature evaluation, APSO for dynamic and optimal feature subset selection, and a DNN that could effectively capture non-linear patterns in the data. The synergistic combination of these components has enabled the proposed model to generalize effectively, while maintaining high discriminative power. The empirical results substantiated the superiority of the proposed IG-APSO-DNN model over existing approaches, demonstrating its potential as a highly effective and reliable solution for anomaly detection tasks.

To provide a more comprehensive comparative perspective, the performance of the proposed IG-APSO-DNN model was further analyzed in relation to the underlying mechanisms of the benchmark methods. The superior results can be attributed to the model's capability to adaptively identify discriminative features through IG and APSO, thereby enhancing representational quality and reducing feature redundancy. In contrast to conventional models such as GCN-LSTM and MFDA-ARRNN-DGRU, which depend on fixed feature extraction strategies, the proposed approach employs dynamic optimization to achieve improved generalization and stability. This analytical comparison reinforces the effectiveness and robustness of the IG-APSO-DNN framework for anomaly detection applications. The superior performance of the proposed IG-APSO-DNN model can be attributed to its integrated framework, where IG efficiently identifies the most discriminative features, and APSO dynamically selects an optimal subset, reducing redundancy and enhancing representational quality. Coupled with the DNN's ability to capture complex non-linear patterns, this synergy enables the model to achieve higher accuracy, balanced precision-recall, and robust generalization compared to conventional approaches that rely on fixed feature extraction or shallow architectures.

The comparative analysis of the False Alarm Rate (FAR), as presented in Fig. 10, further substantiates the effectiveness of the proposed IG-APSO-DNN model in reducing false positive detections, as a critical requirement in any IDS. As depicted, the IG-APSO-DNN model achieved the lowest FAR of 2.11 %, demonstrating a significant improvement over other state-of-the-art method. The A-BiTG and AT-GVAE reported FARs of 5.45 % and 4.79 %, respectively, while the DAMGAT achieved a more

Table 9
The classification report of the proposed GI-APSO-DNN model.

	precision	recall	F1-score	support
0	0.99	0.98	0.98	27,257
1	0.97	0.98	0.97	18,887
accuracy			98	46,144
macro avg	0.98	0.98	0.98	46,144
weighted avg	0.98	0.98	0.98	46,144

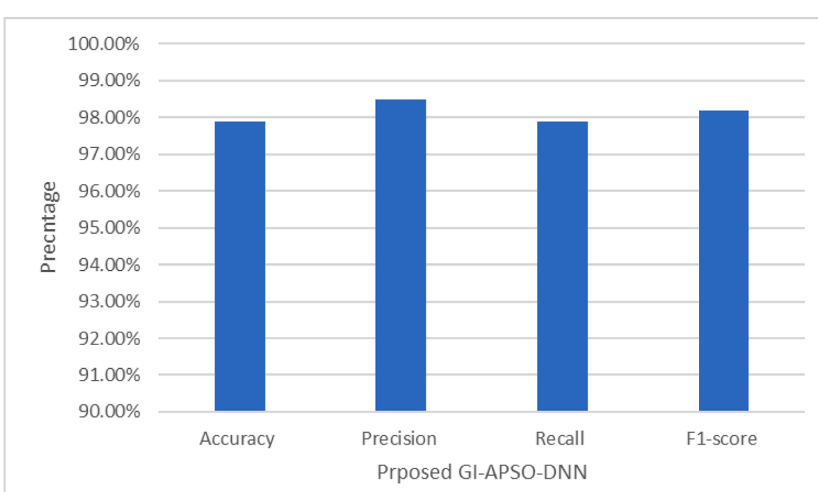

Fig. 6. The overall performance of the proposed GI-APSO-DNN model.

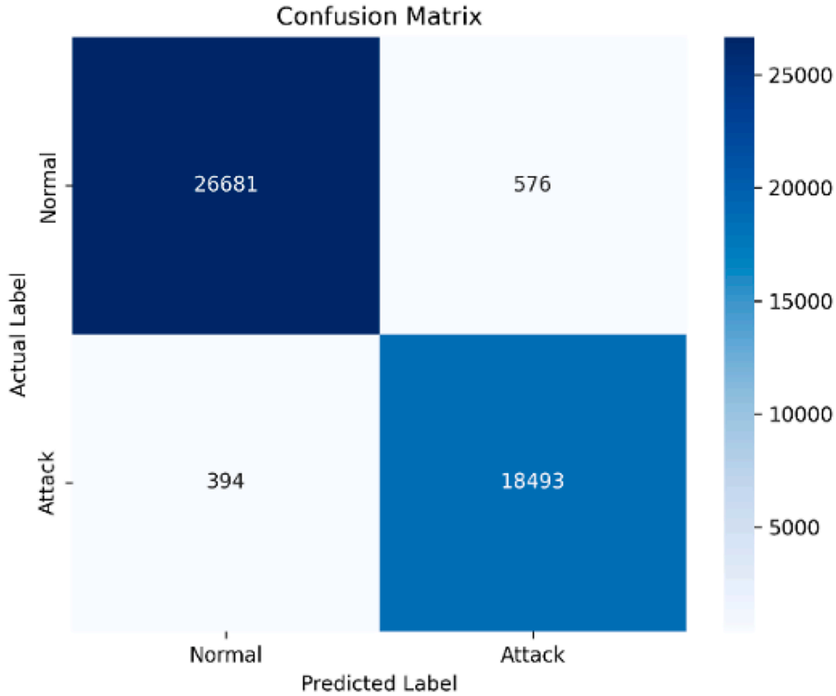

Fig. 7. The confusion matrix of the proposed GI-APSO-DNN model.

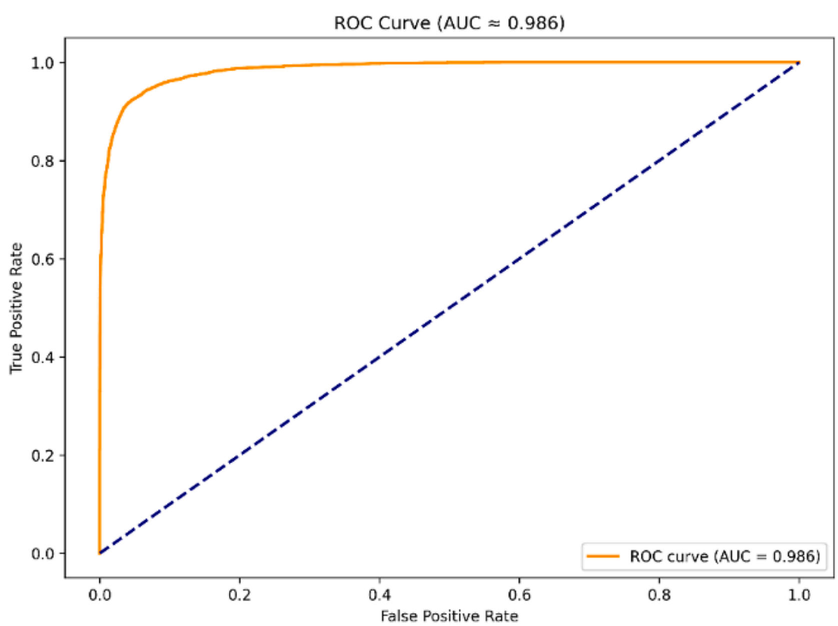

Fig. 8. The ROC_Curve of the proposed GI-APSO-DNN model.

competitive rate of 3.35 %. In contrast, the MFDA-ARRNN-DGRU model exhibited the highest FAR of 5.97 %, indicating a higher incidence of false alarms, which may hinder its operational efficiency. The FAR values of the DAE and the GCN-LSTM were not explicitly reported in the original studies, precluding a direct numerical comparison for these models. The notable reduction in the FAR value achieved by the proposed IG-APSO-DNN highlighted its superior ability to minimize false positives, thereby enhancing its applicability and reliability in real-world security-sensitive environments. This improvement could be attributed to the integrated optimization of relevant features using IG and APSO to filter out noisy or redundant attributes, which allowed the DNN classifier to operate on highly informative inputs.

6. Limitations and future work

The IG-APSO-DNN framework demonstrates strong performance in FDIA detection within smart grids, achieving high accuracy and efficiency. However, several limitations may affect its deployment in real-world scenarios. First, while the current DNN architecture is relatively lightweight, the combined use of IG and APSO for feature selection can introduce additional computational overhead, which may impact real-time implementation in large-scale or resource-constrained environments. Second, the model has been evaluated primarily on the ICS Cyber Attack Dataset, and its performance across diverse operational conditions, varying grid topologies, or unseen attack types remains to be fully

Table 10
Comparison between the proposed GI-APSO-DNN and existing IDS models.

Paper	Model	Accuracy	Precision	Recall	F1-Score	FAR
He, et al. [32]	A-BiTG	93.72 %	91.79 %	92.64	92.12 %	5.45 %
Wang, et al. [49]	AT-GVAE	93.53 %	92.54 %	90.95 %	91.74 %	4.79 %
Su, et al. [50]	DAMGAT	94.88 %	94.71 %	92.16 %	93.42 %	3.35 %
Wang, et al. [48]	DAE	94.10 %	99.60 %	88.60 %	93.80 %	–
Scholar [27]	MFDA-ARRNN-DGRU	94.03 %	94.02 %	94.03 %	94.03 %	5.97 %
Wang, et al. [29]	GCN-LSTM	95.88 %	93.10 %	90.23 %	91.64 %	–
Proposed model	IG-APSO-DNN	97.90 %	98.50 %	97.90 %	98.20 %	2.11 %

validated. Third, like many deep learning approaches, interpretability is limited, which could hinder operator trust and regulatory compliance. Future work will address these limitations by exploring adaptive tuning of IG and APSO parameters to reduce computational demands, extending validation through cross-validation, adversarial scenarios, and real-world datasets, and integrating interpretability tools to provide transparency. Additionally, investigating hybrid optimization strategies and scalable architectures will help ensure that the framework remains robust and effective across evolving cyber-physical environments while maintaining its current strengths in accuracy, reliability, and detection efficiency.

7. Conclusion

An innovative two-stage anomaly detection framework called the IG-APSO-DNN is proposed in this paper as a solution to the essential problem of identifying False Data Injection Attacks (FDIAs) in smart grid environments. An optimized feature selection approach that used IG and APSO was incorporated into the framework, along with a DNN classifier that successfully captured both spatial and temporal dependencies in smart grid data. The addition of IG and APSO not only decreased the dimensionality of the data, but also considerably improved the performance of the detection model by filtering out redundant or noisy

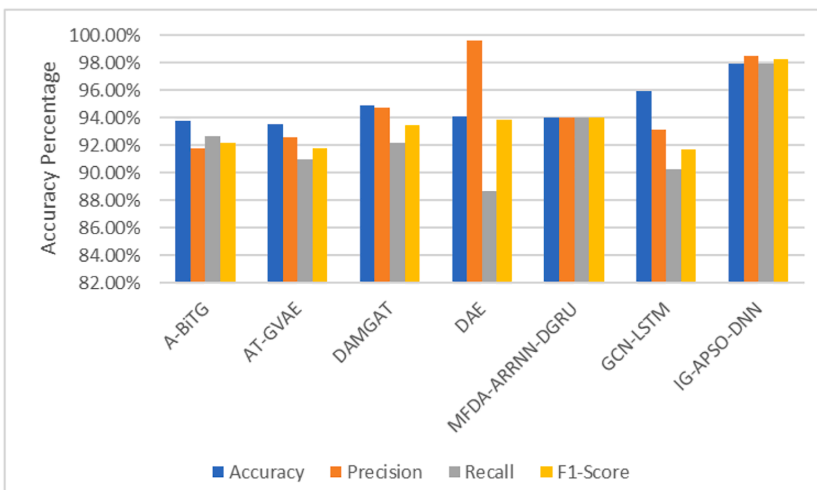

Fig. 9. Comparison between the proposed GI-APSO-DNN and existing IDS models.

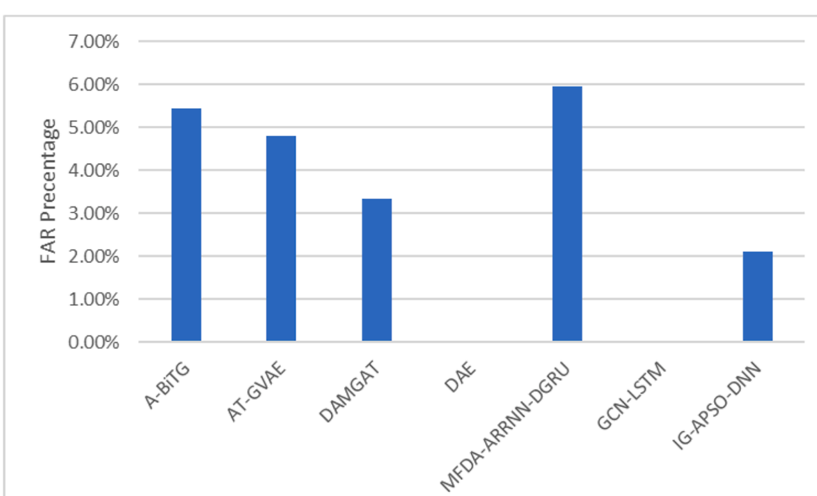

Fig. 10. FAR Performances of Existing Methods and the Proposed IG-APSO-DNN Model.

features. Through the empirical evaluation with the ICS Cyber Attack Dataset, which could simulate realistic FDIA scenarios, this study has demonstrated that the proposed IG-APSO-DNN model outperformed existing state-of-the-art approaches using a number of different performance metrics. These metrics included accuracy, precision, recall, F1-score, and False Alarm Rate (FAR). Notably, the proposed model attained a higher accuracy of 97.90 % and the lowest FAR of 2.11 %, which reflected its excellent capability in differentiating legitimate data from harmful intrusions with little false positives. This was a noteworthy achievement. This improvement could be due to the integrated optimization of the model, as well as its ability to learn discriminative features, which were critical for accurate anomaly identification in complex cyber-physical systems. For the purpose of improving the cybersecurity posture of smart grids, the proposed IG-APSO-DNN offers a solution that is both robust and scalable. By providing enhanced detection accuracy and operating efficiency, it could successfully solve the constraints of classic intrusion detection systems (IDSs) and the deep learning technologies that are already in use. Future work could concentrate on validating this framework in real-world smart grid deployments by incorporating adaptive online learning mechanisms, exploring hybrid or ensemble optimization algorithms, and addressing scalability and real-time processing challenges that could guarantee broad applicability in a variety of industrial contexts.

CRediT authorship contribution statement

Saad Hammood Mohammed: Writing – original draft, Methodology, Investigation, Conceptualization. **Mandeep S. Jit Singh:** Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Abdulmajeed Al-Jumaily:** Formal analysis, Conceptualization. **Mohammad Tariqul Islam:** Software, Resources, Project administration, Funding acquisition. **Md. Shabiul Islam:** Validation, Resources, Project administration, Funding acquisition. **Abdulmajeed M. Alenezi:** Writing – review & editing, Visualization. **Mohamad A. Alawad:** Writing – review & editing, Visualization, Validation. **Muaadh A. Alsoufi:** Writing – review & editing, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no conflicts of interest to report regarding the present study.

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Data availability

No data was used for the research described in the article.

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